

Product Requirements Document

AI-Powered Independent Study Platform

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1. Executive Summary

The platform aims to become a single AI-powered study ecosystem for Indian learners, unifying separate applications for competitive exams, college learning, concept revision, and research understanding into one product.

It supports aspirants preparing for government examinations, students studying academic subjects, and professionals or researchers who want to understand complex material faster.

2. Vision

Become the default study operating system for 500 million learners by combining AI-powered planning, personalized learning, instant assessments, adaptive revision, and performance analytics.

3. Goals

- Provide support for all major Indian competitive examinations.
- Enable instant quiz generation from any topic, PDF, article, YouTube transcript, or research paper.
- Create personalized study plans based on exam, available time, strengths, and weaknesses.
- Track learning progress and automatically recommend revisions.
- Offer a consistent experience across web and mobile.

4. Problem Statement

Students rely on multiple disconnected platforms. Competitive exam aspirants switch between different apps for notes, mocks, current affairs, and planning.

Academic learners have no reliable way to verify understanding of a lecture or research paper.

Most systems promote passive learning rather than active recall and personalized practice.

5. Target Users

- **Government Exam Aspirants:** Preparing for UPSC, SSC, Banking, Railway, Defence, State PSC, JEE, NEET, GATE and similar exams.
- **School & College Students:** Learning syllabus, semester subjects, placements, or board examinations.
- **Researchers & Professionals:** Understanding research papers, standards, documentation, and technical concepts.

6. Product Scope

The platform consists of these major modules:

- Authentication & User Profiles
- Exam Explorer with syllabus database
- Academic Learning Hub
- AI Study Plan Generator
- Instant Quiz Generator
- Mock Test Engine
- Research Paper & PDF Learning
- Schedule and Calendar
- Performance Analytics
- Revision Engine
- Recommendation Engine
- Notifications and Reminders
- Subscription & Payments
- Admin Dashboard

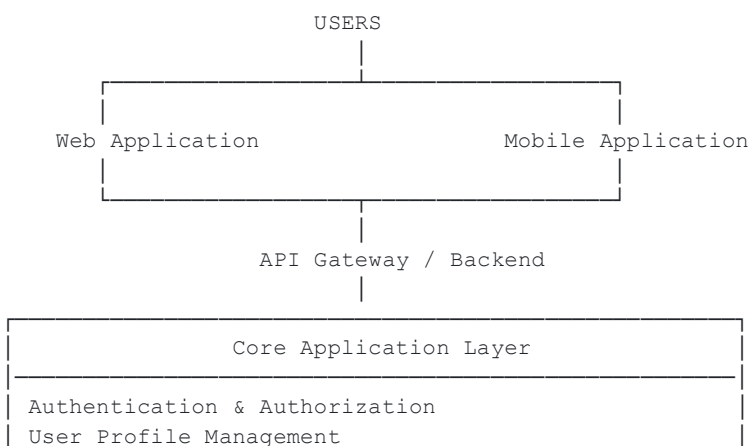
6.1 System Architecture Overview

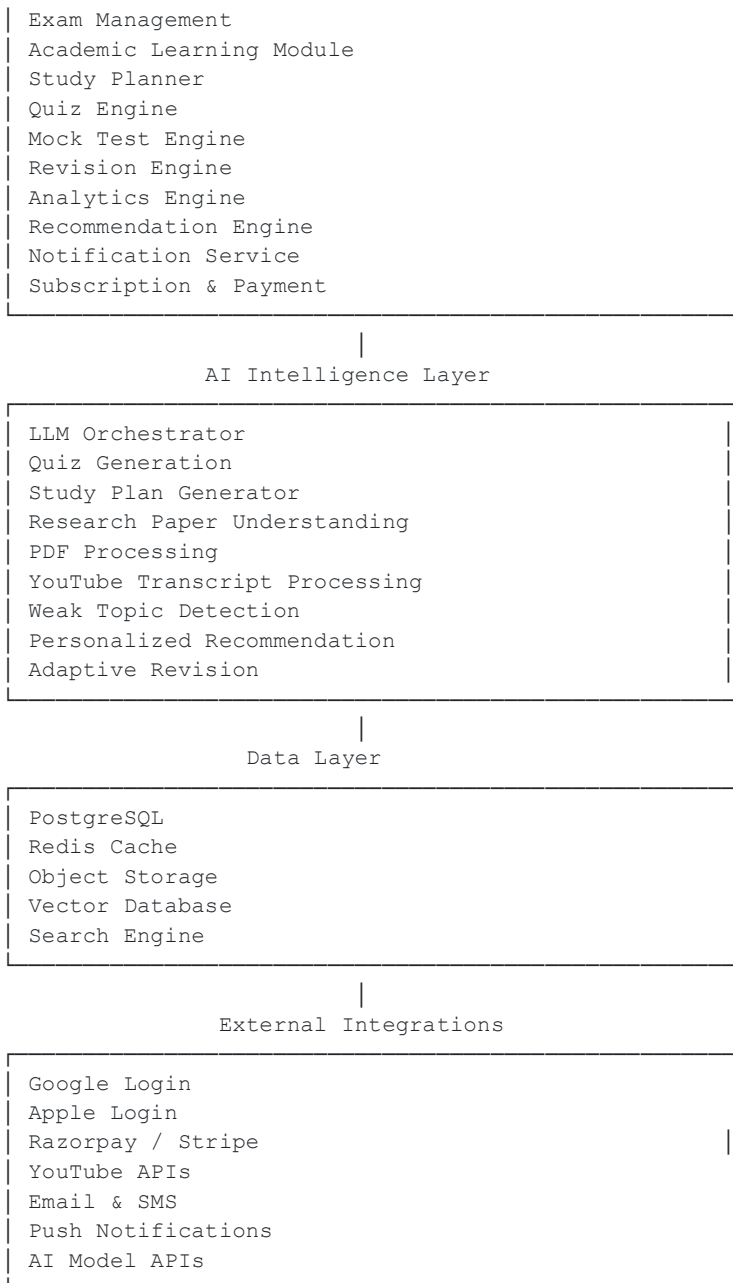
Overview

The AI Learning Operating System follows a cloud-native, modular, service-oriented architecture designed for scalability, maintainability, and future AI integration. The system separates the presentation layer, business logic, AI services, data layer, and third-party integrations into independent components, allowing each module to scale and evolve without affecting the rest of the platform.

The architecture is built around an AI-first philosophy, where intelligent services enhance planning, learning, assessment, recommendations, and analytics while the core application manages user interactions, content, and system workflows.

Architecture Diagram (Textual Representation)





Major Components

Presentation Layer: Provides responsive interfaces for Web and Mobile users. Responsible for user interaction, navigation, content display, accessibility, and real-time updates.

Backend Services: Acts as the application's central orchestration layer. Responsibilities include Authentication, User Management, Exam Management, Course Management, AI Request Routing, Subscription Management, Progress Tracking, and Analytics Collection.

AI Intelligence Layer: The AI layer powers the platform's intelligent features. Major capabilities include AI-generated quizzes, personalized study plans, weak concept identification, PDF understanding,

research paper summarization, YouTube transcript analysis, adaptive revision scheduling, personalized recommendations, and performance prediction.

Data Layer: Stores structured and unstructured data. Includes user profiles, exams, syllabus, study plans, quiz history, analytics, uploaded PDFs, AI embeddings, and learning history.

Third-party Services: The system integrates with external services for authentication, AI models, payments, notifications, video processing, and communication.

Scalability Considerations

- Millions of concurrent learners
- Thousands of examinations
- Millions of quizzes
- AI model upgrades
- Multi-language support
- Future institutional deployments

7. Core Functional Requirements

7.1 Exam Engine

Allows users to search, select, and enroll in any supported examination, providing syllabus, study roadmap, previous-year papers, and mock tests.

- Search and filter by exam name, category, or conducting body.
- View detailed syllabus with topic-wise breakdown.
- Access curated study roadmap with milestone timelines.
- Download and attempt previous-year question papers.

7.2 Instant Quiz Engine

Allows users to paste a topic, upload a PDF, provide a YouTube link, or choose a syllabus topic, and AI generates quizzes of varying difficulty with explanations.

- Input sources: text, PDF, YouTube URL, topic selection.
- Difficulty levels: Easy, Medium, Hard.
- Question types: Multiple choice, true/false, fill-in-the-blank.
- AI generates explanations for correct and incorrect answers.

7.3 Study Planner

Creates personalized daily and weekly plans that automatically adapt when progress changes.

- Inputs: Exam date, daily available hours, strengths/weaknesses.
- Outputs: Daily task list with time blocks.
- Auto-adjusts when user misses sessions or improves accuracy.

7.4 Analytics

Overview

Analytics is the intelligence backbone of the platform. Instead of merely displaying scores, the analytics engine continuously evaluates learning behavior, predicts future performance, identifies weaknesses, and recommends actionable improvements.

Analytics operate at three levels: Learner Analytics, Content Analytics, and Platform Analytics.

Learner Analytics

Each learner receives a personalized dashboard containing:

- Performance Metrics
 - Overall Accuracy
 - Topic-wise Accuracy
 - Subject-wise Accuracy
 - Exam-wise Performance
 - Average Quiz Score
 - Mock Test Score
 - Rank Prediction
 - Percentile
 - Learning Velocity
 - Completion Percentage
- Learning Behaviour
 - Daily Study Time
 - Weekly Consistency
 - Monthly Progress
 - Study Streak
 - Active Days
 - Preferred Study Hours
 - Learning Speed
- Concept Mastery

Every concept receives a mastery score. Example visual classification:

Photosynthesis:  95%

Genetics:  72%

Electrochemistry:  51%

Concepts are classified as: Mastered, Strong, Moderate, Weak, or Critical.

- Weak Area Detection

- Frequently incorrect topics
- Slow-solving concepts
- Low-confidence answers
- Forgotten concepts
- Long-unrevised topics

The system generates a personalized revision list automatically.

- Recommendation Engine

- Next topic to study
- Videos
- Articles
- Practice sets
- Revision schedule
- Mock tests
- Daily goals

Instructor/Admin Analytics

Administrators can monitor:

- Daily Active Users (DAU) & Monthly Active Users (MAU)
- Quiz Generation Count & AI Usage
- Subscription Conversion & Revenue
- Retention Rate & User Drop-off
- Popular Exams & Popular Subjects
- System Health

AI Insights

- "Your Physics accuracy has improved by 18% this month."
- "Organic Chemistry requires immediate revision."
- "Your predicted JEE percentile is 96.4."
- "Completing three additional mock tests may improve your score by approximately 4–6%."

7.5 Revision

Automatically collects incorrectly answered questions and schedules spaced revision.

- Spaced repetition algorithm based on Ebbinghaus curve.
- Daily revision reminders for due items.
- Progress tracking for revision mastery.

7.6 Recommendations

Suggest videos, articles, practice sets, and next topics based on learning history.

- Collaborative filtering from user cohort data.
- Content-based filtering from topic similarity.
- In-app notification for recommended resources.

8. Non-Functional Requirements

- **Responsive UI:** Seamless experience across desktop, tablet, and mobile.
- **Scalable architecture:** Cloud-native design with auto-scaling.
- **Security:** Secure authentication (OAuth 2.0) and encrypted data at rest and in transit.
- **Performance:** Low response latency (2 seconds for quiz generation, 500ms for typical page loads).
- **Maintainability:** Modular microservice-friendly design with clear API contracts.
- **Accessibility:** WCAG 2.1 AA compliance.

9. User Journeys

9.1 Government Exam Aspirant Journey

1. Register and create profile.
2. Select exam (e.g., UPSC CSE).
3. AI creates personalized study plan.
4. Daily study with topic quizzes and mock tests.
5. View analytics dashboard.
6. Adaptive revision of weak topics.
7. Receive readiness score before exam.

9.2 Academic Learner Journey

1. Choose subject or upload content (PDF or YouTube link).
2. AI summarizes key concepts.
3. Generate quiz to verify understanding.
4. Identify weak concepts from results.
5. Personalized revision materials suggested.
6. Track mastery over time.

9.3 User Flow Diagram

A. Competitive Exam Aspirant Flow

Open Application -> Login / Register -> Create Profile -> Select Competitive Exam

-> AI Generates Personalized Study Plan -> Dashboard -> [Study / Practice / Mock Test]
-> Performance Analytics -> Weak Topic Detection -> AI Revision Plan -> Daily Progress Update -> Exam Ready

B. Student / Academic Learner Flow

Open Application -> Login -> Select Subject -> Choose Learning Source (Topic, PDF, Research Paper, YouTube)
-> AI Content Processing -> Summary Generation -> Instant Quiz -> Performance Analysis
-> Concept Mastery Score -> Personalized Revision -> Knowledge Mastery

C. Premium Subscription Flow

Free User -> Uses Daily Free Limits -> Limit Reached -> Upgrade Prompt -> Subscription Selection -> Payment Gateway -> Payment Success -> Premium Features Activated

10. AI Capabilities

- Quiz generation from diverse content formats.
- PDF content understanding and extraction.
- YouTube transcript parsing and concept mapping.
- Research paper summarization and key point extraction.
- Adaptive study planning based on user data.
- Weak-topic detection through performance analytics.
- Personalized learning recommendations.
- Progress prediction and readiness forecasting.

11. Success Metrics

Key Performance Indicators (KPIs)

Metric	Target (Month 1)	Target (Month 6)
Daily Active Users (DAU)	10,000	100,000
Quiz Completion Rate	70%	85%
Study Plan Adherence	40%	65%
Average Accuracy	10%	20%

Improvement (30 days)		
Retention (Day 30)	50%	60%
Subscription Conversion Rate	5%	12%

12. MVP (Minimum Viable Product)

The first production release includes:

- Authentication and user profiles.
- Exam selection with syllabus browser.
- AI-generated study plans.
- Topic and PDF-based quiz generation.
- Performance analytics dashboard.
- Revision engine with spaced repetition.
- Push notifications and reminders.

Future phases will include: collaborative learning, community discussions, multilingual AI tutors, and institution dashboards.

13. Future Roadmap

- Voice AI tutor for hands-free learning.
- Offline mode for uninterrupted study.
- Institution dashboards for educators.
- Coaching center integrations.
- Interview preparation module (HR and technical).
- AI mentor with conversational Q&A.
- Collaborative study rooms and group challenges.
- Multilingual support covering all major Indian languages.